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KOSH-AI

NSAP Scheme Demand Predictor

An AI-Powered District-Level Classification and Decision-Support System using IBM watsonx.ai

Project Report

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Executive Summary

Kosh-AI is a district-level machine learning system that predicts the most likely National Social Assistance Programme (NSAP) pension scheme pattern for a given district, using demographic and beneficiary statistics published under the AI Kosh open data initiative. The project was developed as part of the Edunet Foundation – IBM Skills Build AICTE Internship Program 2026, and demonstrates a complete, cloud-native machine learning lifecycle using IBM watsonx.ai and IBM Watson Machine Learning — from raw data ingestion through automated model training to a live, cloud-hosted REST API.

The National Social Assistance Programme provides financial assistance to elderly citizens, widows, and persons with disabilities from economically vulnerable households, distributed across three constituent schemes: the Indira Gandhi National Old Age Pension Scheme (IGNOAPS), the Indira Gandhi National Widow Pension Scheme (IGNWPS), and the Indira Gandhi National Disability Pension Scheme (IGNDPS). Administering this at scale, across hundreds of districts and multiple states, involves considerable manual review effort and is susceptible to inconsistency. Kosh-AI addresses this by learning the association between a district's demographic composition — gender distribution, social-category representation, and digital-identity linkage — and the scheme pattern that dominates in that district, using the AI Kosh dataset as training data.

The system was built using IBM watsonx.ai AutoAI, which automated data preprocessing, feature handling, pipeline generation, and hyperparameter optimization across a set of candidate ensemble classifiers. The champion pipeline — a Snap Random Forest Classifier accelerated by IBM's SnapML library — was selected based on holdout evaluation, refined further through incremental batch training, registered in the IBM watsonx.ai model repository, and deployed as an online scoring service through IBM Watson Machine Learning. A live scoring request against this deployed endpoint was used to confirm the system functions correctly end-to-end.

Kosh-AI is explicitly scoped as a decision-support tool, not an eligibility-determination system. It operates on aggregated district-level data and produces scheme-pattern predictions intended to assist welfare planning and resource allocation; it does not, and is not intended to, replace the statutory, rule-based process by which individual citizens are certified eligible for a specific NSAP scheme.

Project at a Glance

Attribute	Details
Domain	Public Welfare / Government Technology
Problem Type	Multiclass Classification
Dataset	AI Kosh NSAP Dataset
Granularity	District Level
Platform	IBM watsonx.ai / IBM Cloud
AutoML	IBM watsonx.ai AutoAI
Selected Model	Snap Random Forest Classifier (Pipeline_5)
Deployment	IBM Watson Machine Learning — Online Deployment
Interface	REST API

Chapter 1 — Introduction

1.1 Background

Data-driven governance increasingly relies on structured, machine-readable records to plan and evaluate public welfare programmes. The National Social Assistance Programme is one of the Government of India's principal social security schemes, administered through state and district-level bodies and split into distinct pension categories — old age, widow, and disability — each with its own eligibility basis and beneficiary profile. District-level planning matters because demographic composition, and therefore welfare demand, varies substantially across India's 700-plus districts; a resource-allocation approach calibrated to national averages can under- or over-serve specific regions.

1.2 Problem Context

Welfare administrations managing NSAP work with large volumes of beneficiary data spread across states and districts. Manually reviewing this data to understand which scheme categories are most active in which districts, or to flag districts that may need additional outreach or resourcing, is labour-intensive and does not scale well as data volume grows. District-level variation in gender ratio, social-category composition, and digital-identity (Aadhaar/mobile) linkage further complicates manual pattern recognition, since the relationship between these factors and scheme demand is not always visually obvious from raw tables.

1.3 Motivation

Machine learning is well suited to this kind of pattern-recognition task precisely because it operates as a decision-support layer rather than a decision-making authority. A trained classifier can surface the dominant scheme pattern implied by a district's demographic profile far faster than manual review, while leaving the actual, statutory eligibility determination for individual beneficiaries to the relevant government authority. This separation of pattern prediction from legal determination is the central design principle behind Kosh-AI, and is revisited in depth in Chapter 9.

1.4 Project Vision

Kosh-AI's broader vision is to demonstrate that an enterprise-grade AI pipeline — spanning cloud storage, automated machine learning, model governance, and live API deployment — can be built end-to-end using accessible tools (IBM Cloud Lite, watsonx.ai AutoAI) and applied to a genuine public-sector dataset, while maintaining a disciplined, responsible-AI boundary between prediction and legal decision-making.

Conceptual Flow

Government Welfare Data → AI Analysis → Scheme Pattern Prediction → Decision Support

Chapter 2 — Problem Statement & Objectives

The National Social Assistance Programme consists of multiple pension schemes, each designed for a different beneficiary category. Manual verification and classification of records at scale is time-consuming, resource-intensive, and prone to inconsistency across districts and states. This project formulates the challenge as a supervised multiclass classification problem:

Formal Problem Definition

Input
District-level demographic and beneficiary statistics (gender distribution, social-category composition, Aadhaar and mobile-number linkage, total beneficiary counts)

Output
Most likely NSAP scheme pattern for the district, expressed as the target variable schemecode

The three possible output classes are:

Code	Scheme
IGNOAPS	Indira Gandhi National Old Age Pension Scheme
IGNWPS	Indira Gandhi National Widow Pension Scheme
IGNDPS	Indira Gandhi National Disability Pension Scheme

Input → Model → Output
District Demographic Profile → Snap Random Forest Classifier (AutoAI-selected) → Predicted schemecode + Class Probabilities

Project Objectives

- Develop an AI-powered classification model for district-level NSAP scheme-pattern prediction.
- Automate the scheme-pattern classification process using machine learning.
- Reduce manual effort involved in welfare planning and scheme-allocation review.
- Improve the consistency and efficiency of welfare-distribution decision-support.
- Demonstrate an end-to-end AI deployment workflow using IBM Cloud.
- Enable real-time predictions through a cloud-hosted REST API.

Chapter 3 — Dataset & Data Understanding

3.1 Dataset Source

The dataset was obtained from AI Kosh, India's national platform for AI-ready datasets maintained under IndiaAI, and reflects official NSAP administrative statistics.

3.2 Dataset Characteristics

Characteristic	Value
Total Records	2,156
Districts Covered	726
States / Union Territories	36
Scheme Categories	3 (IGNOAPS, IGNWPS, IGNDPS)
Financial Year	2025–2026
Granularity	District-level, aggregated per scheme

3.3 Features

Column	Description
finyear	Financial year of the record
lgdstatecode	Local Government Directory state code
statename	State / Union Territory name
lgddistrictcode	Local Government Directory district code
districtname	District name
totalbeneficiaries	Total beneficiaries recorded under the scheme
totalmale / totalfemale / totaltransgender	Beneficiaries by gender
totalsc / totalst / totalgen / totalobc	Beneficiaries by social category
totalaadhaar	Beneficiaries with Aadhaar linked
totalmobilenumber	Beneficiaries with mobile number linked

3.4 Target Variable

The model target is schemecode, a categorical field identifying which NSAP scheme the aggregated record belongs to.

3.5 Target Classes

Code	Scheme
IGNOAPS	Indira Gandhi National Old Age Pension Scheme

Code	Scheme
IGNWPS	Indira Gandhi National Widow Pension Scheme
IGNDPS	Indira Gandhi National Disability Pension Scheme

3.6 Dataset Composition Overview

Figure 2 — Structural Summary

2,156 records = 726 districts × up to 3 scheme categories, all drawn from a single financial year (2025–2026) and reported at district-level aggregation rather than individual-applicant level.

Figure 3 — Feature Category Groupings

Identifiers (finyear, state/district codes and names) | Beneficiary Volume (totalbeneficiaries) | Demographic Composition (gender, social category) | Digital Inclusion (Aadhaar, mobile linkage)

No numerical class-distribution or feature-distribution statistics beyond the counts above are asserted here, in keeping with the requirement to report only figures explicitly verified from the project's source notebooks and README.

Chapter 4 — System Architecture

The end-to-end Kosh-AI pipeline moves data from its open-data source through cloud storage, automated model development, and into a live deployment layer. The high-level flow is summarized below, followed by the two architecture diagrams produced for this project.

> **Pipeline Sequence**
> AI Kosh Dataset → IBM Cloud Object Storage → IBM watsonx.ai Studio → Data Preparation → AutoAI Experiment → Pipeline Generation → Model Evaluation → Snap Random Forest Selection → Incremental Training → Model Registration → IBM Deployment Space → Online Deployment → REST API → Real-Time Prediction

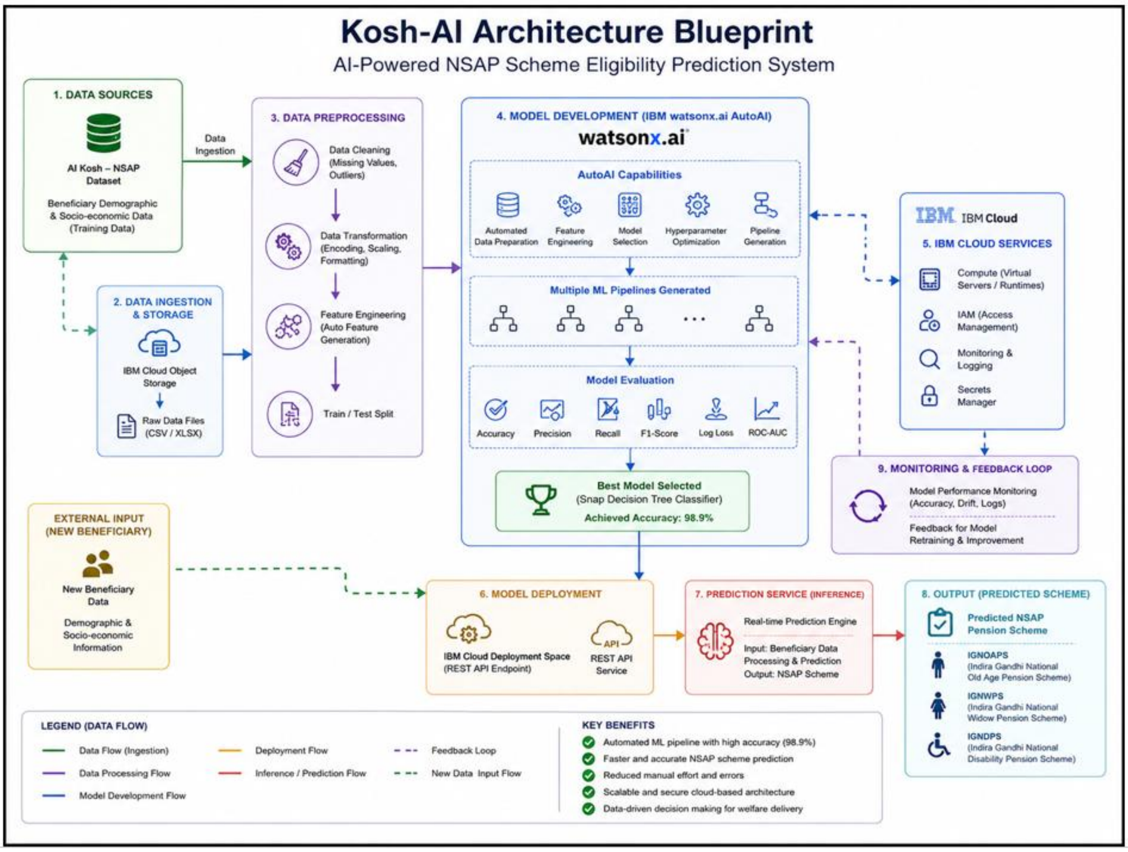

Figure 4. Kosh-AI Architecture Blueprint — end-to-end data, AutoAI, and cloud deployment flow.

> **Note on Figure 4 labelling**
> This diagram was produced during early project planning and labels the selected model as "Snap Decision Tree Classifier" with an accuracy of 98.9%. The verified pipeline confirmed in the project notebooks is the Snap Random Forest Classifier (Pipeline_5); no overall model accuracy figure has been independently verified — the only confirmed metric is a 98.67% class probability on a single live test prediction (see Chapter 7). The architectural flow shown in the diagram remains accurate; the model's name and accuracy label should be read against the corrected values used throughout this report.

Figure 5. Kosh-AI System Workflow — stage-by-stage view from dataset to scheme prediction.

Note on Figure 5 labelling

As with Figure 4, this workflow diagram references "Snap Decision Tree Classifier" and a 98.9% accuracy figure predating the corrected pipeline identification. Read the stage sequence as accurate; read the model name as Snap Random Forest Classifier and treat the accuracy figure as unverified.

Chapter 5 — Methodology

5.1 Data Ingestion

The NSAP dataset was uploaded to IBM Cloud Object Storage as a project data asset and accessed within IBM watsonx.ai Studio through its Cloud Object Storage integration, loading the data as a pandas DataFrame for the AutoAI experiment.

5.2 Data Preparation

AutoAI performed automated preprocessing on the ingested data, including handling of categorical fields and preparation of the feature set ahead of pipeline generation, as configured in the experiment.

5.3 AutoML Configuration

Parameter	Value
Prediction type	Multiclass classification
Prediction column	schemecode
Scoring metric	Accuracy
Holdout size	10%

5.4 Pipeline Generation

AutoAI evaluated multiple batched ensemble pipelines under its include_batched_ensemble_estimators configuration, covering Snap Random Forest, Batched Extra Trees, and Batched LightGBM variants — all accelerated through IBM's SnapML library.

5.5 Model Selection

Candidate pipelines were ranked by holdout accuracy using the AutoAI pipeline optimizer summary. The top-ranked pipeline, Pipeline_5 — a Snap Random Forest Classifier — was selected as the champion model and extracted as a scikit-learn-compatible estimator for further use.

5.6 Incremental Training

A dedicated notebook resumed Pipeline_5 as a lale-wrapped estimator and continued training through batch-wise partial_fit() calls across the dataset, with learning-curve tracking across batches. Incremental training of this kind is valuable because it allows the champion pipeline to absorb new financial-year data as it becomes available, without needing to be rebuilt from scratch through a fresh AutoAI run.

5.7 Model Registration

The trained pipeline was registered in the IBM watsonx.ai project's model repository, giving it a governed, versioned identity ahead of deployment.

5.8 Deployment

A dedicated IBM Deployment Space was created to separate production assets from development-stage resources. The registered model was promoted into this space and deployed as an Online Deployment, exposing a secured REST scoring endpoint for real-time inference.

Chapter 6 — Machine Learning Model

The champion pipeline selected by AutoAI for this project is a Snap Random Forest Classifier, an ensemble-of-decision-trees algorithm implemented through IBM's SnapML library and exposed within the pipeline as a BatchedTreeEnsembleClassifier.

How the model works, in this project's context

- Ensemble learning: rather than relying on a single model, the classifier trains many individual decision trees and combines their outputs, which typically improves robustness over any one tree in isolation.
- Decision trees: each tree splits the district-level feature set (gender ratios, social-category composition, digital-linkage rates) into branching decision rules that ultimately map to one of the three scheme classes.
- Randomized feature selection: individual trees in the forest are trained on randomized subsets of features and data, which reduces correlation between trees and helps limit overfitting to any one pattern in the training set.
- Batched training and SnapML acceleration: SnapML's batched implementation allows the ensemble to be trained and updated in data batches, which is what enabled the incremental partial_fit() training described in Chapter 5.

Within Kosh-AI specifically, this architecture fits the project's needs because it (a) handles a moderate number of numeric demographic features without requiring extensive manual feature engineering, (b) supports the incremental-update workflow needed for future financial-year data, and (c) produces calibrated class-probability outputs, which is what the deployed REST API returns alongside the predicted class. This report does not claim Snap Random Forest to be superior to other algorithms in general terms — only that it was the AutoAI-selected champion for this specific dataset and configuration.

Conceptual Model Flow

Multiple Decision Trees → Ensemble Aggregation → Class Prediction → Probability Vector

Chapter 7 — Model Evaluation & Results

AutoAI evaluated the following pipelines during the experiment. Pipeline_5 was selected as the deployed champion based on holdout evaluation.

Pipeline	Model	Status
Pipeline_5	Snap Random Forest Classifier	Selected
Other pipeline	Batched Extra Trees Classifier	Evaluated
Other pipeline	Batched LightGBM Classifier	Evaluated

The evaluation and deployment process followed these stages: holdout evaluation of all generated pipelines; ranking of pipelines by holdout accuracy through the AutoAI pipeline optimizer; selection of the top-ranked pipeline; extraction of feature importances for the champion pipeline; incremental batch training via partial_fit(); and, finally, validation of the deployed endpoint through a live scoring call.

Live Prediction Result

The following result was returned by the deployed online endpoint for a live scoring request:


```
[
  {
    "fields": ["prediction", "probability"],
    "values": [
      [
        "IGNDPS",
        [0.9867446884372569, 0.013255311562742558, 0]
      ]
    ]
  }
]
```


Prediction Result Card

Predicted Scheme Code: IGNDPS — Indira Gandhi National Disability Pension Scheme. Class Probability: 98.67% (with 1.33% and ~0% assigned to the remaining two classes).

Important Distinction — Confidence ≠ Accuracy

The 98.67% figure above is the model's probability estimate for one specific test input, not a measure of overall model accuracy across the full dataset. No independently verified overall accuracy metric is reported in this document; only this single-sample class-probability result is confirmed from the project's live API test.

Chapter 8 — API & Cloud Deployment

Moving a trained pipeline from an AutoAI experiment into a callable service involves several governed stages within IBM watsonx.ai and Watson Machine Learning:

> **Deployment Sequence**
> Model → Model Repository → Deployment Space → Online Deployment → REST Endpoint → Scoring Request → Prediction + Probability

Registering the model creates a versioned artifact inside the watsonx.ai project. Promoting it into a Deployment Space isolates production-facing assets from the experimental project workspace. Creating an Online Deployment then exposes that artifact as a REST endpoint that accepts a JSON payload of district-level features and returns a predicted schemecode with its associated probability vector, as shown in Chapter 7. This REST interface is what allows Kosh-AI to serve real-time predictions to any authorized client application, rather than requiring predictions to be generated inside a notebook session.

Example Scoring Request (illustrative, credentials omitted)

```
POST /ml/v4/deployments/<deployment_id>/predictions
Authorization: Bearer <token>
Content-Type: application/json

{
 "input_data": [{
   "fields": ["lgdstatecode", "statename", "districtname",
             "totalbeneficiaries", "totalmale", "totalfemale",
             "totalsc", "totalst", "totalgen", "totalobc",
             "totalaadhaar", "totalmobilenumber"],
   "values": [[1, "JAMMU AND KASHMIR", "ANANTNAG", ...]]
 }]
}
```

Chapter 9 — Responsible AI & Project Scope

This section defines, deliberately and explicitly, what Kosh-AI is and is not designed to do.

WHAT KOSH-AI DOES	WHAT KOSH-AI DOES NOT DO
Predicts district-level NSAP scheme demand/pattern from aggregated demographic data, to support welfare planning and resource allocation.	Determine individual citizen eligibility for any NSAP scheme. Eligibility remains a statutory, rule-based determination made by the relevant government authority.

Discussion

- Human oversight: predictions are intended to inform, not replace, human decision-makers in welfare administration.
- Fairness and bias: because the training data includes SC/ST/OBC and gender composition, model behaviour across these subgroups should be audited before any operational use, to check that prediction errors are not concentrated in specific demographic groups.
- Explainability: feature-importance outputs from the AutoAI pipeline provide a starting point for interpretability, but a dedicated explainability layer (see Chapter 10) would be needed for full auditability of individual predictions.
- Data limitations: the dataset reflects a single financial year (2025–2026) at district-level aggregation; it does not contain individual applicant records, income data, or historical multi-year trends.
- Aggregation limitations: because records are aggregated per district and scheme, the model cannot observe or predict anything about a specific person; its unit of prediction is the district-scheme record.
- Decision-support vs. automated decision-making: Kosh-AI is positioned strictly as the former.

Chapter 10 — Limitations & Future Enhancements

Current Limitations

- Single financial-year dataset — no historical trend data is currently incorporated.
- District-level aggregation only — no individual-applicant granularity.
- No independently verified overall accuracy metric beyond the single live-prediction confidence score reported in Chapter 7.
- No explainability (e.g., SHAP/LIME) layer implemented yet.
- No fairness/bias audit has yet been performed across demographic subgroups.

Future Roadmap

Phase	Enhancement
Phase 1	Current system — AutoAI-trained Snap Random Forest, deployed via REST API
Phase 2	Multi-year time-series modelling as further financial-year data becomes available
Phase 3	Explainable AI (SHAP / LIME) for transparent, auditable predictions
Phase 4	Fairness and bias audits across gender and social-category subgroups
Phase 5	Interactive analytics dashboard (IBM Cognos or Streamlit)
Phase 6	Geospatial visualization of scheme distribution across districts
Phase 7	CI/CD-driven automated retraining (GitHub Actions / IBM Continuous Delivery)
Phase 8	Integration with additional government welfare datasets

Chapter 11 — Conclusion

Kosh-AI demonstrates a complete machine learning lifecycle applied to a genuine public-sector dataset: ingestion of open district-level NSAP data into IBM Cloud Object Storage, automated pipeline generation and model selection through IBM watsonx.ai AutoAI, incremental refinement of the champion Snap Random Forest pipeline, governed model registration, and deployment as a live, queryable REST API through IBM Watson Machine Learning.

Technically, the project shows that AutoML tooling does not remove the need for engineering judgment — data ingestion configuration, AutoML scoring and holdout choices, pipeline evaluation, incremental-training design, model registration, and deployment-space management all required deliberate decisions, even though AutoAI automated the underlying pipeline search itself.

From a public-sector relevance standpoint, the project illustrates how a district-level welfare dataset can support planning-oriented decision support without overstepping into individual eligibility determination — a distinction maintained deliberately throughout this report and central to how such a system should be scoped in any real deployment context.

Kosh-AI, in its current form, is a demonstrated AI decision-support architecture rather than a production system operating within an actual government department. Its future scalability lies in incorporating multi-year data, explainability, fairness auditing, and CI/CD-driven retraining, as outlined in Chapter 10.

Closing Statement
Kosh-AI illustrates the role of AI as a decision-support layer for data-driven governance — augmenting, not replacing, statutory human decision-making in public welfare administration.

References & Acknowledgements

Technologies & Platforms Referenced

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